CI/CD Pipeline For Node.js Retail Application

Course Name: Devops

Institution Name: Medicaps University – Datagami Skill Based Course

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Group Name: Group 01D5

Project Number: DO-01

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University Mentor Name: Prof. Ritesh Joshi

Academic Year: 4th

Problem Statement & Objectives

1. Problem Statement

In modern software development, deploying applications manually is time-consuming and prone to errors. It becomes difficult to maintain consistency, reliability, and speed in software delivery.

The problem addressed in this project is to design and implement a Continuous Integration and Continuous Deployment (CI/CD) pipeline for a Node.js-based retail application.

The system uses modern DevOps tools such as:

- GitHub for version control
- GitHub Actions for automation
- Docker for containerization
- Kubernetes for deployment
- Amazon Web Services for cloud hosting

The objective is to automate the entire software delivery lifecycle to ensure fast, reliable, and efficient deployment.

2. Project Objectives

- To automate code integration and deployment process
- To reduce manual errors in software deployment
- To ensure continuous testing and integration
- To deploy the application in a containerized environment
- To host the application on cloud infrastructure
- To improve scalability and reliability of the system

3. Scope of the Project

The scope of this project includes:

i. Development of a simple Node.js retail web application
ii. Implementation of CI/CD pipeline using GitHub Actions
iii. Docker-based containerization of the application
iv. Deployment using Kubernetes cluster
v. Hosting the application on AWS EC2 instance
vi. Providing a working frontend with product listing and cart system

This project focuses on demonstrating a complete DevOps lifecycle from development to deployment.

Proposed Solution

The proposed solution is an automated CI/CD pipeline system where:

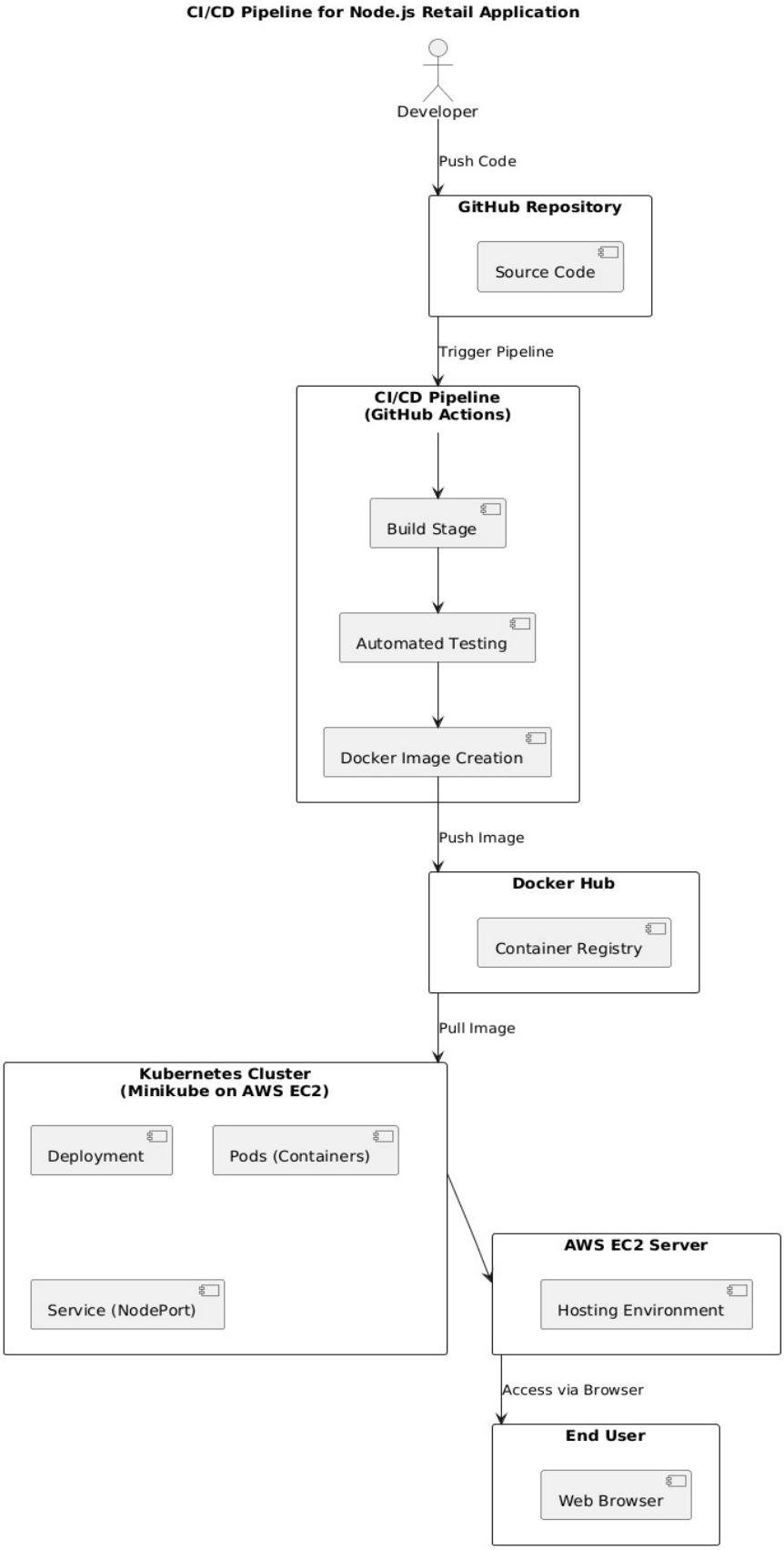
- Developer writes and pushes code to GitHub
- GitHub Actions automatically triggers pipeline
- Application is tested and built
- Docker image is created
- Image is pushed to Docker Hub
- Kubernetes pulls the image and deploys containers
- Application is hosted on AWS EC2

This solution ensures:

- Faster deployments
- High availability
- Reduced human intervention
- Continuous delivery

1. Key features
    - Automated CI/CD pipeline
    - Real-time deployment on cloud
    - Docker-based containerized application
    - Kubernetes orchestration
    - Node.js REST API
    - Interactive frontend retail dashboard
    - Product listing and cart system

2. Overall Architecture / Workflow



The system follows a structured workflow:

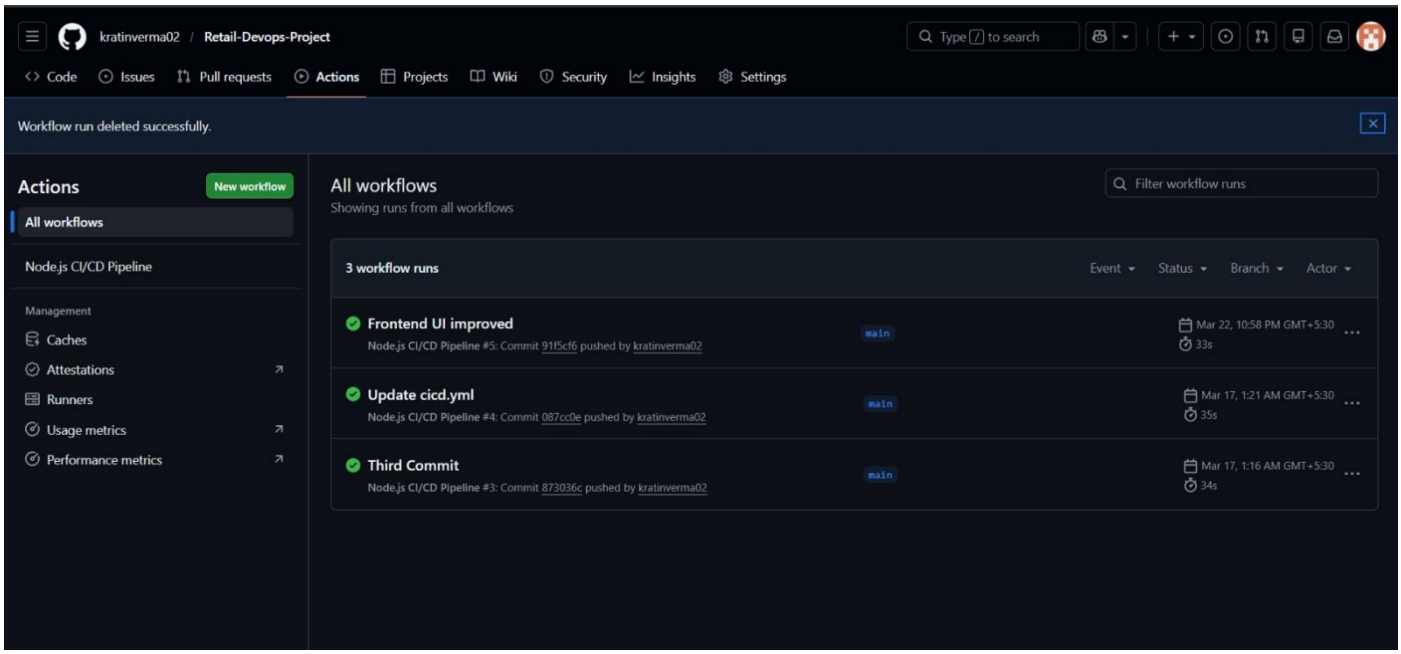
i. Developer pushes code to GitHub repository
ii. GitHub Actions triggers CI/CD pipeline
iii. Pipeline performs build and testing
iv. Docker image is created
v. Image is pushed to Docker Hub
vi. Kubernetes pulls image and creates pods
vii. Application runs inside containers
viii. Hosted on AWS EC2 server

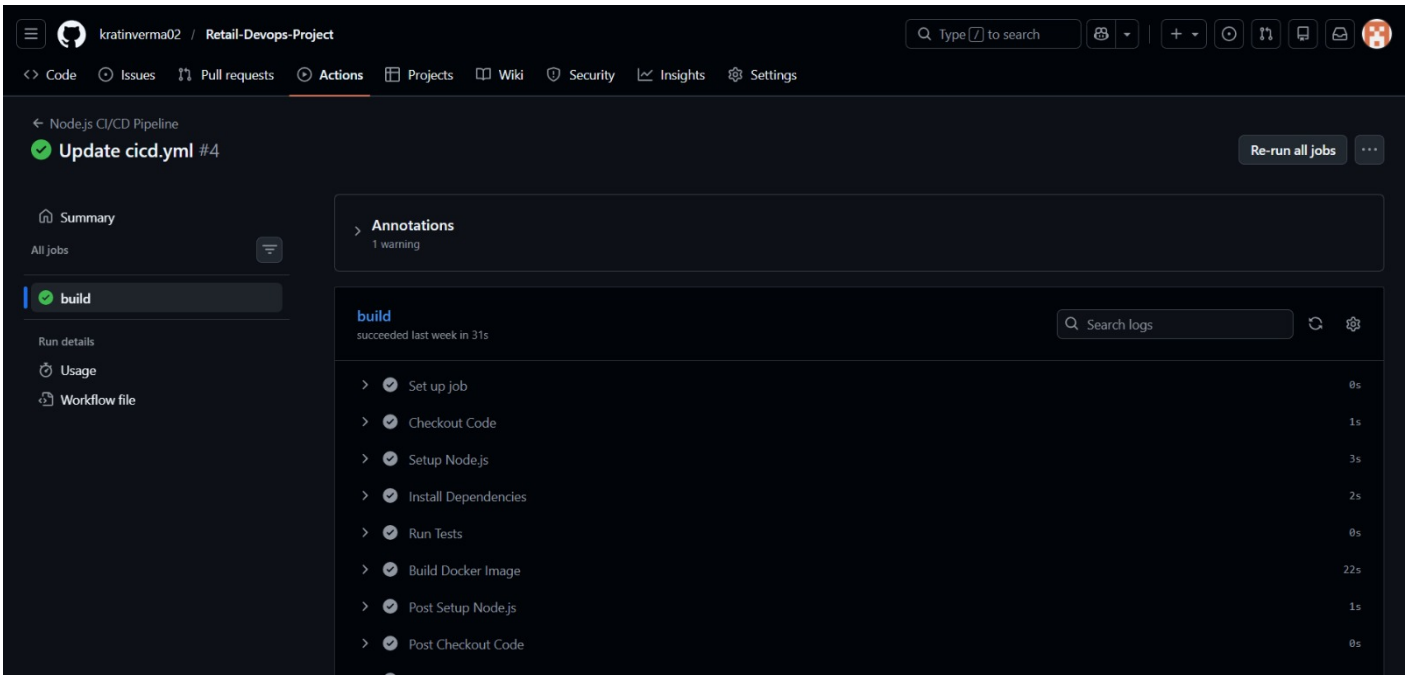
3. Tools & Technologies Used (*If applicable)*

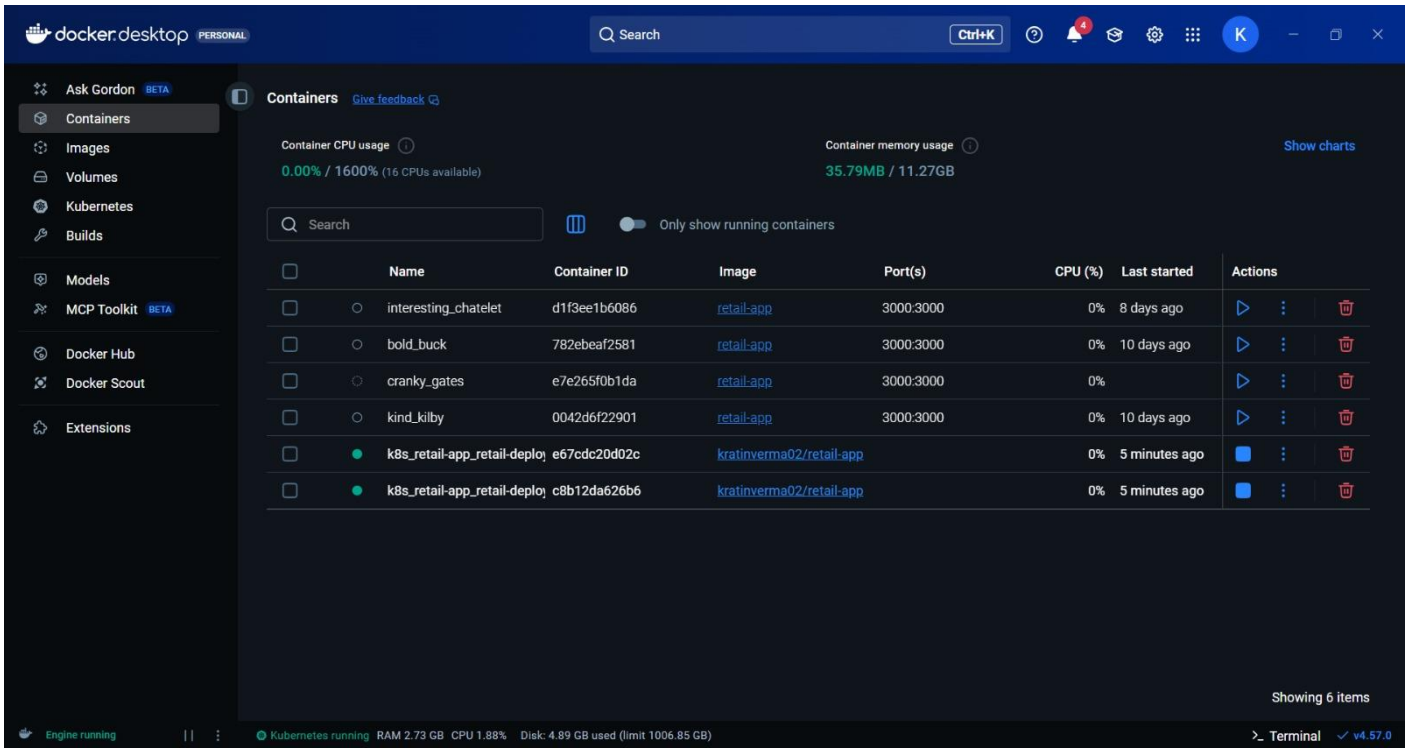
Tool	Description
Git	Tracks code changes
GitHub	Repository hosting
GitHub Actions	CI/CD automation
Node.js	Backend development
Docker	Containerization
Kubernetes	Deployment management
Amazon Web Services	Cloud hosting

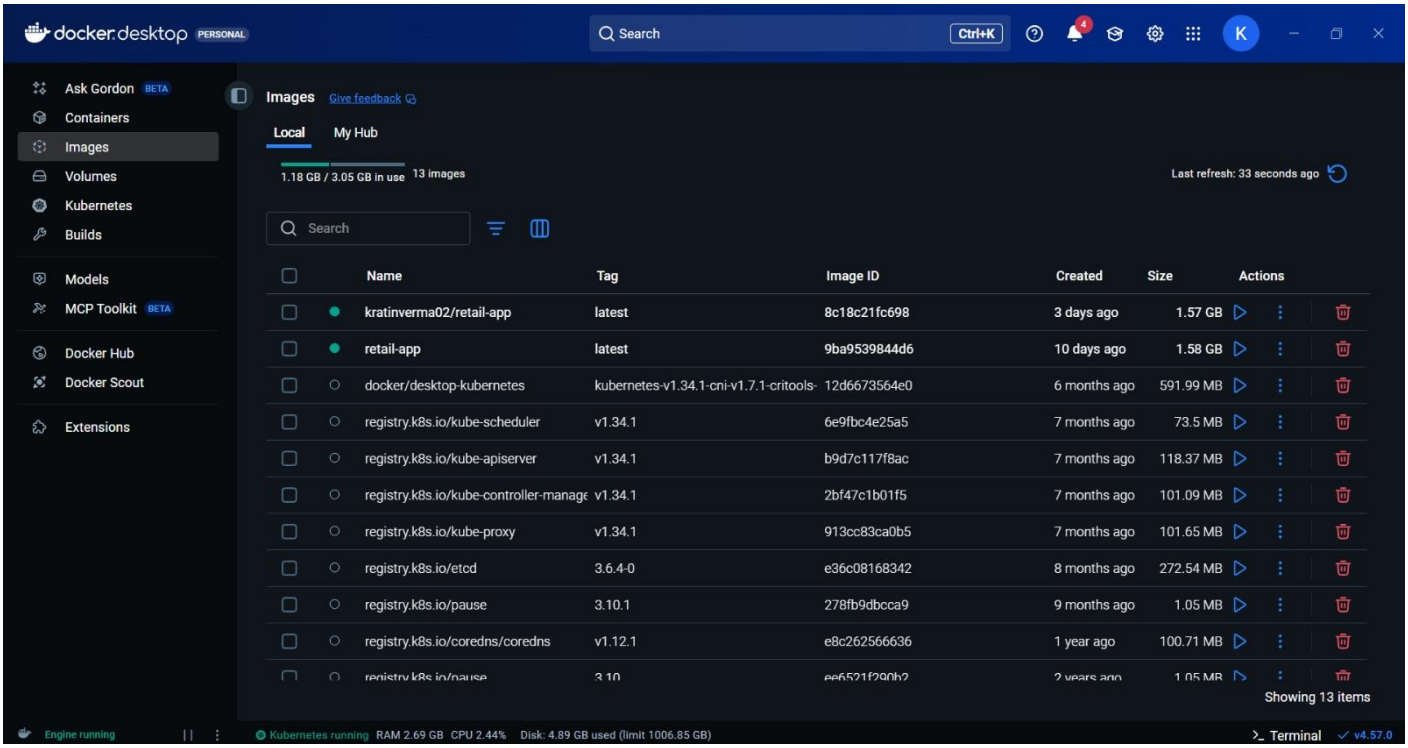
Results & Output

1. *Screenshots / outputs*




kratinverma02 / Retail-Devops-Project
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Node.js CI/CD Pipeline
Update cicd.yml #4
Re-run all jobs
Summary
All jobs
build
Run details
Usage
Workflow file
Annotations
1 warning
build
succeeded last week in 31s
Search logs
Set up job 0s
Checkout Code 1s
Setup Node.js 3s
Install Dependencies 2s
Run Tests 0s
Build Docker Image 22s
Post Setup Node.js 1s
Post Checkout Code 0s


docker desktop PERSONAL
Search Ctrl+K
Ask Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Containers Give feedback
Container CPU usage
0.00% / 1600% (16 CPUs available)
Container memory usage
35.79MB / 11.27GB
Show charts
Search
Only show running containers
Name Container ID Image Port(s) CPU (%) Last started Actions
interesting_chatelet d1f3ee1b6086 retail-app 3000:3000 0% 8 days ago
bold_buck 782ebeaf2581 retail-app 3000:3000 0% 10 days ago
cranky_gates e7e265f0b1da retail-app 3000:3000 0%
kind_kilby 0042d6f22901 retail-app 3000:3000 0% 10 days ago
k8s_retail-app_retail-deplo e67cdc20d02c kratinverma02/retail-app 0% 5 minutes ago
k8s_retail-app_retail-deplo c8b12da626b6 kratinverma02/retail-app 0% 5 minutes ago
Showing 6 items
Engine running
Kubernetes running RAM 2.73 GB CPU 1.88% Disk: 4.89 GB used (limit 1006.85 GB)
Terminal v4.57.0


docker desktop PERSONAL
Search Ctrl+K
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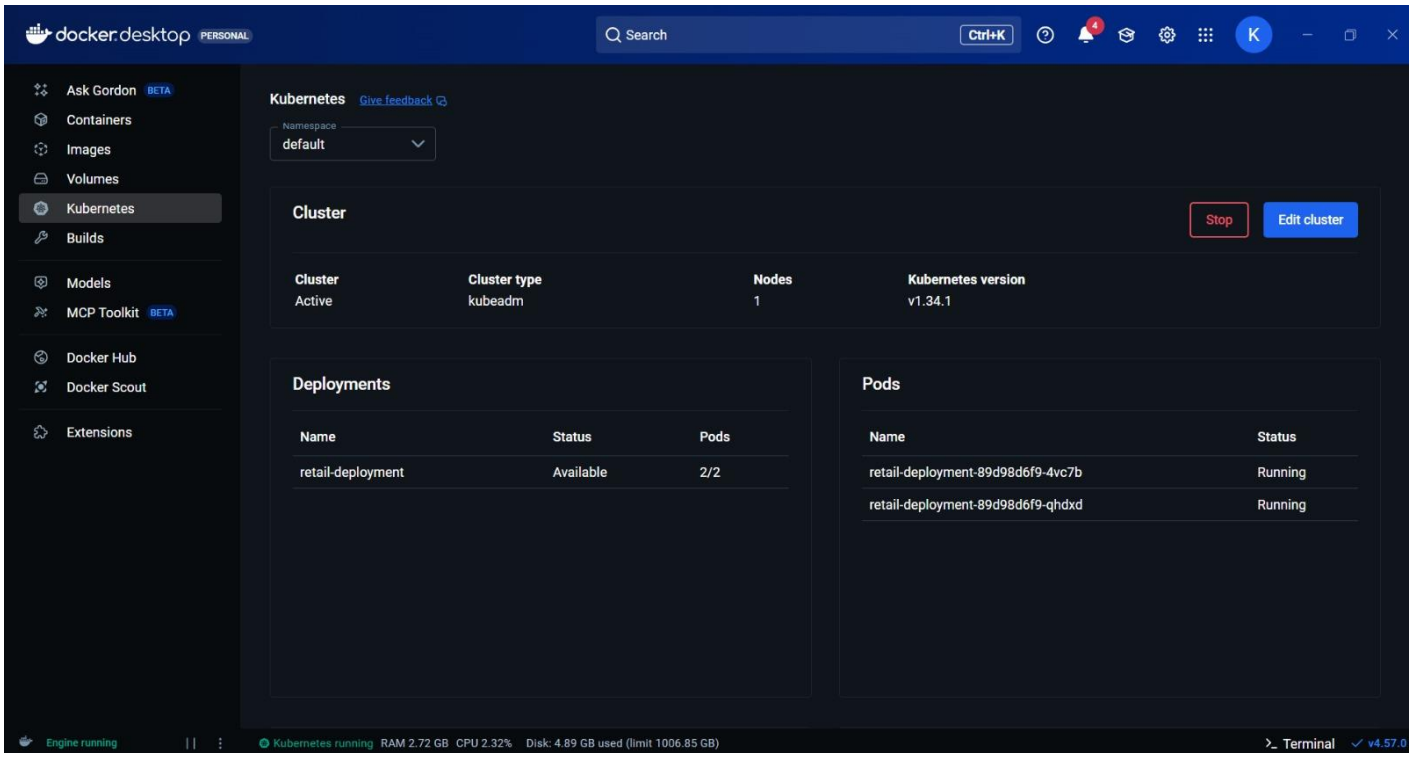
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Images Give feedback
Local My Hub
1.18 GB / 3.05 GB in use 13 images
Last refresh: 33 seconds ago
Search
Name Tag Image ID Created Size Actions
kratinverma02/retail-app latest 8c18c21fc698 3 days ago 1.57 GB
retail-app latest 9ba9539844d6 10 days ago 1.58 GB
docker/desktop-kubernetes kubernetes-v1.34.1-cni-v1.7.1-critools- 12d6673564e0 6 months ago 591.99 MB
registry.k8s.io/kube-scheduler v1.34.1 6e9fbc4e25a5 7 months ago 73.5 MB
registry.k8s.io/kube-apiserver v1.34.1 b9d7c117f8ac 7 months ago 118.37 MB
registry.k8s.io/kube-controller-manage v1.34.1 2bf47c1b01f5 7 months ago 101.09 MB
registry.k8s.io/kube-proxy v1.34.1 913cc83ca0b5 7 months ago 101.65 MB
registry.k8s.io/etcd 3.6.4-0 e36c08168342 8 months ago 272.54 MB
registry.k8s.io/pause 3.10.1 278fb9dbcca9 9 months ago 1.05 MB
registry.k8s.io/coredns/coredns v1.12.1 e8c262566636 1 year ago 100.71 MB
Showing 13 items
Engine running
Kubernetes running RAM 2.69 GB CPU 2.44% Disk: 4.89 GB used (limit 1006.85 GB)
Terminal v4.57.0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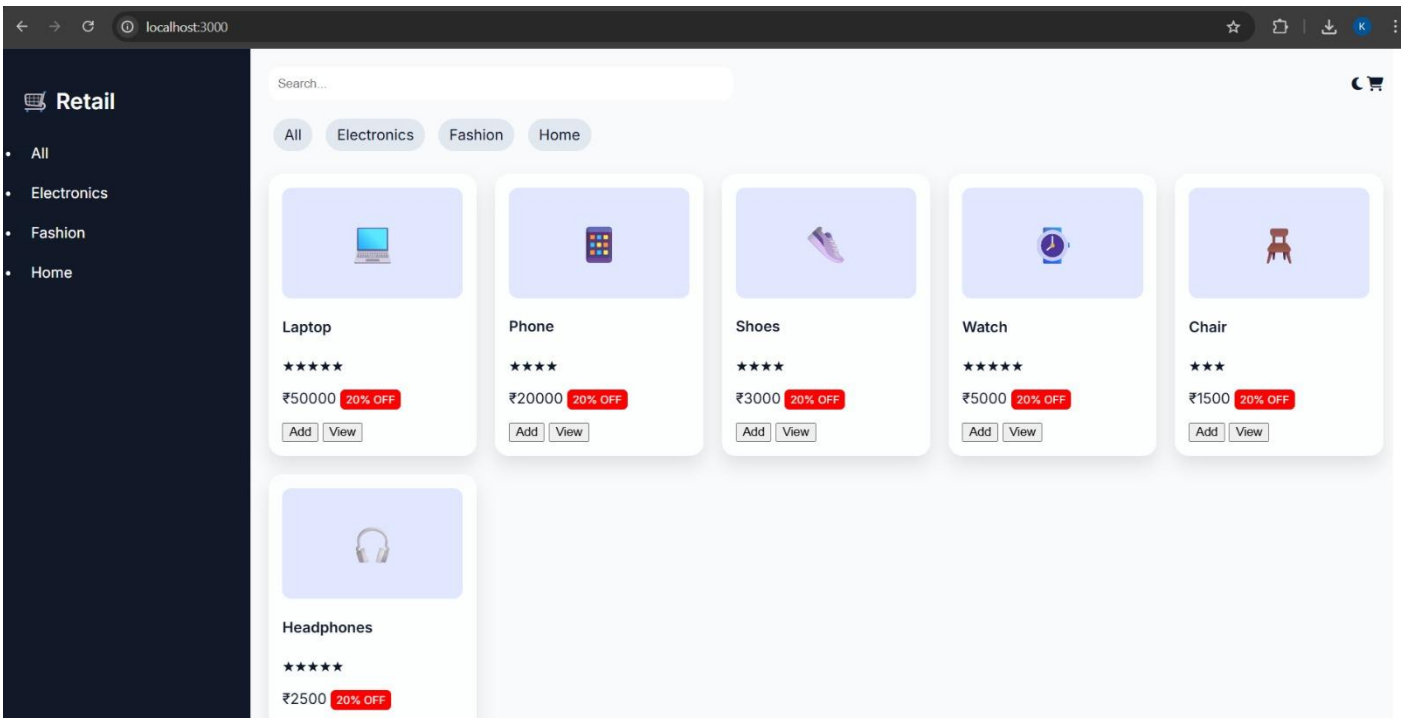
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Kubernetes Give feedback
Namespace
default
Cluster
Stop
Edit cluster
Cluster Active
Cluster type kubeadm
Nodes 1
Kubernetes version v1.34.1
Deployments
Name Status Pods
retail-deployment Available 2/2
Pods
Name Status
retail-deployment-89d98d6f9-4vc7b Running
retail-deployment-89d98d6f9-qhdxd Running
Engine running
Kubernetes running RAM 2.72 GB CPU 2.32% Disk: 4.89 GB used (limit 1006.85 GB)
Terminal v4.57.0



2. *Reports / dashboards / models*
   - *CI/CD logs showing build success*
   - *Kubernetes deployment status*
   - *Docker container logs*

3. *Key outcomes*
   - *Successful implementation of CI/CD pipeline*
   - *Automated build and deployment*
   - *Application successfully hosted on AWS*
   - *Reduced deployment time significantly*
   - *Improved system scalability and reliability*

Conclusion

This project demonstrates a complete DevOps implementation by building a CI/CD pipeline for a Node.js retail application.

The integration of GitHub, GitHub Actions, Docker, Kubernetes, and AWS provides a powerful solution for modern software deployment.

The system ensures:

- Automation
- Scalability
- Reliability
- Faster delivery

Thus, the project successfully achieves its objective of automating the software delivery lifecycle.

Future Scope & Enhancements

- Integration of monitoring tools like Prometheus and Grafana
- Auto-scaling using Kubernetes
- Use of Jenkins for advanced CI/CD pipelines
- Database integration (MongoDB / MySQL)
- Improved UI/UX design
- Implement authentication and security features